

# FEMMInterpreter

## Output Class Attributes

### Attributes

#### Top-Level

| Attribute         | Type        | Description                                            |
|-------------------|-------------|--------------------------------------------------------|
| format_version    | float       | Format version of the file.                            |
| frequency_hz      | float       | Frequency for harmonic problems.                       |
| solver_precision  | float       | Computing precision within FEMM.                       |
| min_angle_deg     | float       | Angle restriction for the triangulation.               |
| model_depth       | float       | Typical length in z-direction (for 2D planar).         |
| length_unit       | str         | Unit used for lengths (e.g., 'millimeters', 'inches'). |
| problem_type      | str         | The 2D problem is either planar or axisymmetric.       |
| coordinate_system | str         | Definition of the coordinate system.                   |
| comment_text      | str         | Problem description.                                   |
| vector_x          | list[float] | X-coordinates of the solution points.                  |
| vector_y          | list[float] | Y-coordinates of the solution points.                  |
| vector_a          | list[float] | Magnetic vector potential values at each point.        |

#### Materials

| Attribute            | Type                | Description                                |
|----------------------|---------------------|--------------------------------------------|
| name                 | str                 | Name of the material.                      |
| permeability         | tuple(float, float) | Permeability in x and y directions.        |
| coercive_force       | float               | Coercive force in A/m.                     |
| coercive_angle       | float               | Angle of coercive force in degrees.        |
| current_density      | tuple(float, float) | Real and imaginary current density.        |
| conductivity         | float               | Electrical conductivity in S/m.            |
| lamination_thickness | float               | Lamination thickness in mm.                |
| hysteresis_angle     | float               | Hysteresis angle for AC analysis.          |
| hysteresis_angle_x   | float               | Hysteresis angle in x-direction.           |
| hysteresis_angle_y   | float               | Hysteresis angle in y-direction.           |
| lamination_type      | int                 | Type of lamination.                        |
| lamination_fill      | float               | Lamination fill factor.                    |
| num_strands          | int                 | Number of strands in a stranded conductor. |
| wire_diameter        | float               | Wire diameter in mm.                       |
| bh_curve             | list[tuple]         | B-H curve data points (H, B).              |

## Boundaries

| Attribute      | Type  | Description                                               |
|----------------|-------|-----------------------------------------------------------|
| name           | str   | Name of the boundary condition.                           |
| boundary_type  | int   | Boundary type                                             |
| prescribed_a   | float | Prescribed A value.                                       |
| prescribed_a_x | float | Prescribed A x-component.                                 |
| prescribed_a_y | float | Prescribed A y-component.                                 |
| phase_angle    | float | Phase angle in degrees for AC analysis (Phi).             |
| mixed_c0       | float | Mixed boundary C0 coefficient real part.                  |
| mixed_c0_imag  | float | Mixed boundary C0 coefficient imaginary part.             |
| mixed_c1       | float | Mixed boundary C1 coefficient real part.                  |
| mixed_c1_imag  | float | Mixed boundary C1 coefficient imaginary part.             |
| mu_ssd         | float | Relative permeability for Small Skin Depth boundary (Mu). |
| sigma_ssd      | float | Conductivity in MS/m for Small Skin Depth boundary (Sig). |
| inner_angle    | float | Inner angle for periodic boundary (degrees).              |
| outer_angle    | float | Outer angle for periodic boundary (degrees).              |

## Circuits

| Attribute       | Type  | Description                                  |
|-----------------|-------|----------------------------------------------|
| name            | str   | Name of the circuit.                         |
| total_amps_real | float | Total current (real component) in amps.      |
| total_amps_imag | float | Total current (imaginary component) in amps. |
| circuit_type    | int   | Type of circuit.                             |

## Methods

| Method          | Parameters            | Description                                                         |
|-----------------|-----------------------|---------------------------------------------------------------------|
| point_potential | x: float, y: float    | Returns the magnetic vector potential A at point (x, y).            |
| field_potential | resolution: int = 300 | Returns interpolated potential field as (x_space, y_space, a_grid). |